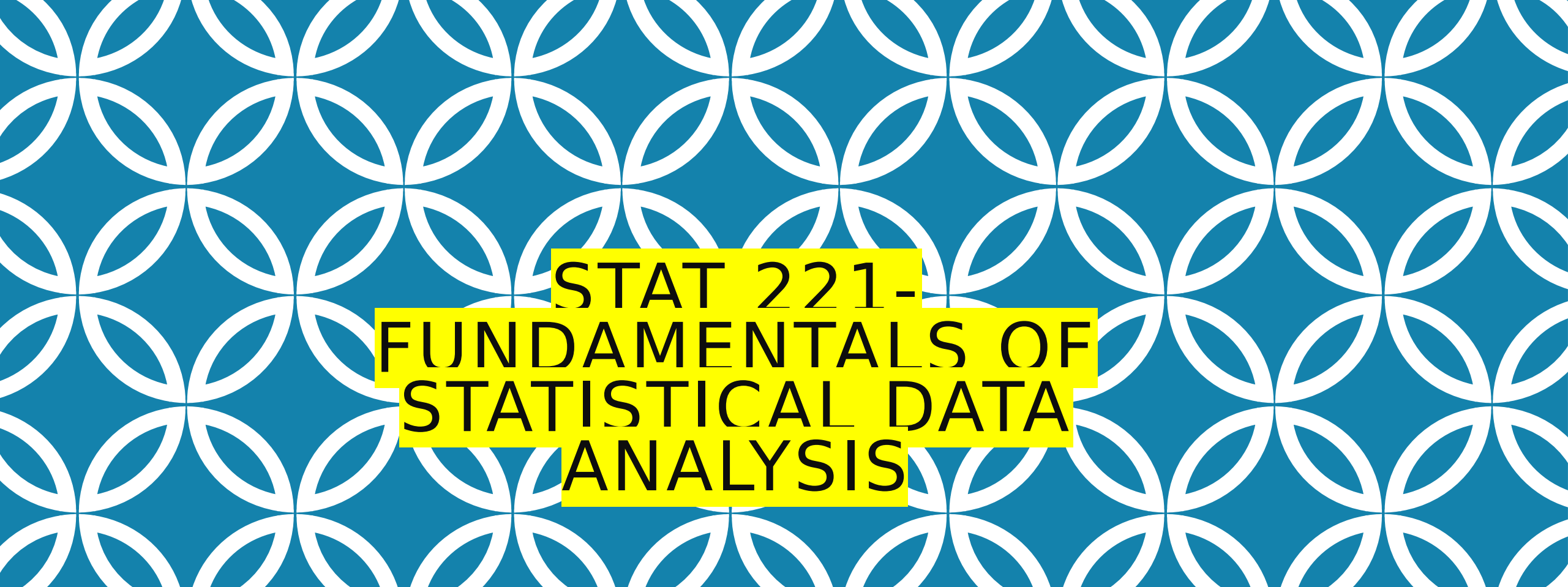


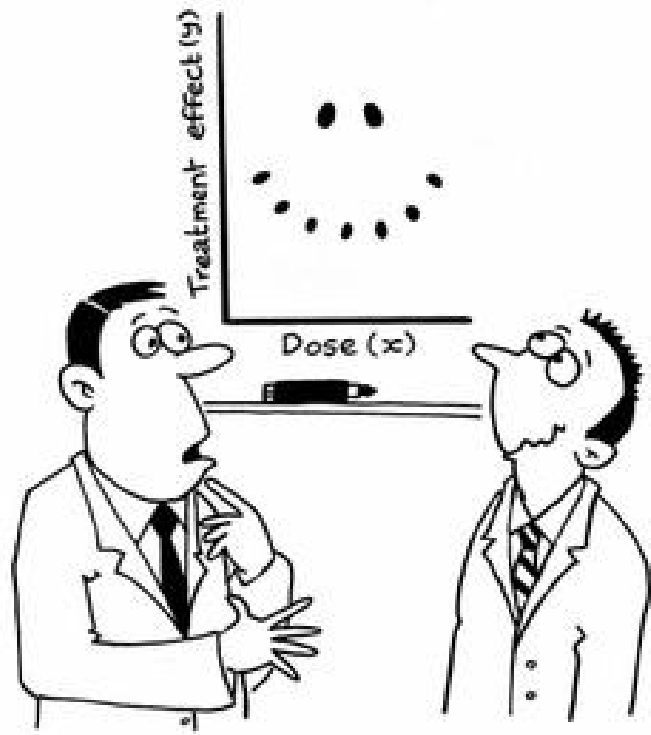


STAT 221- FUNDAMENTALS OF STATISTICAL DATA ANALYSIS

Week-6: Measures of Dispersion;
Distributions

Shivakumar Jolad

Shivakumar Jolad
Associate
Professor, FLAME
University, Pune



"It's a non-linear pattern with outliers.....but for some reason I'm very happy with the data."

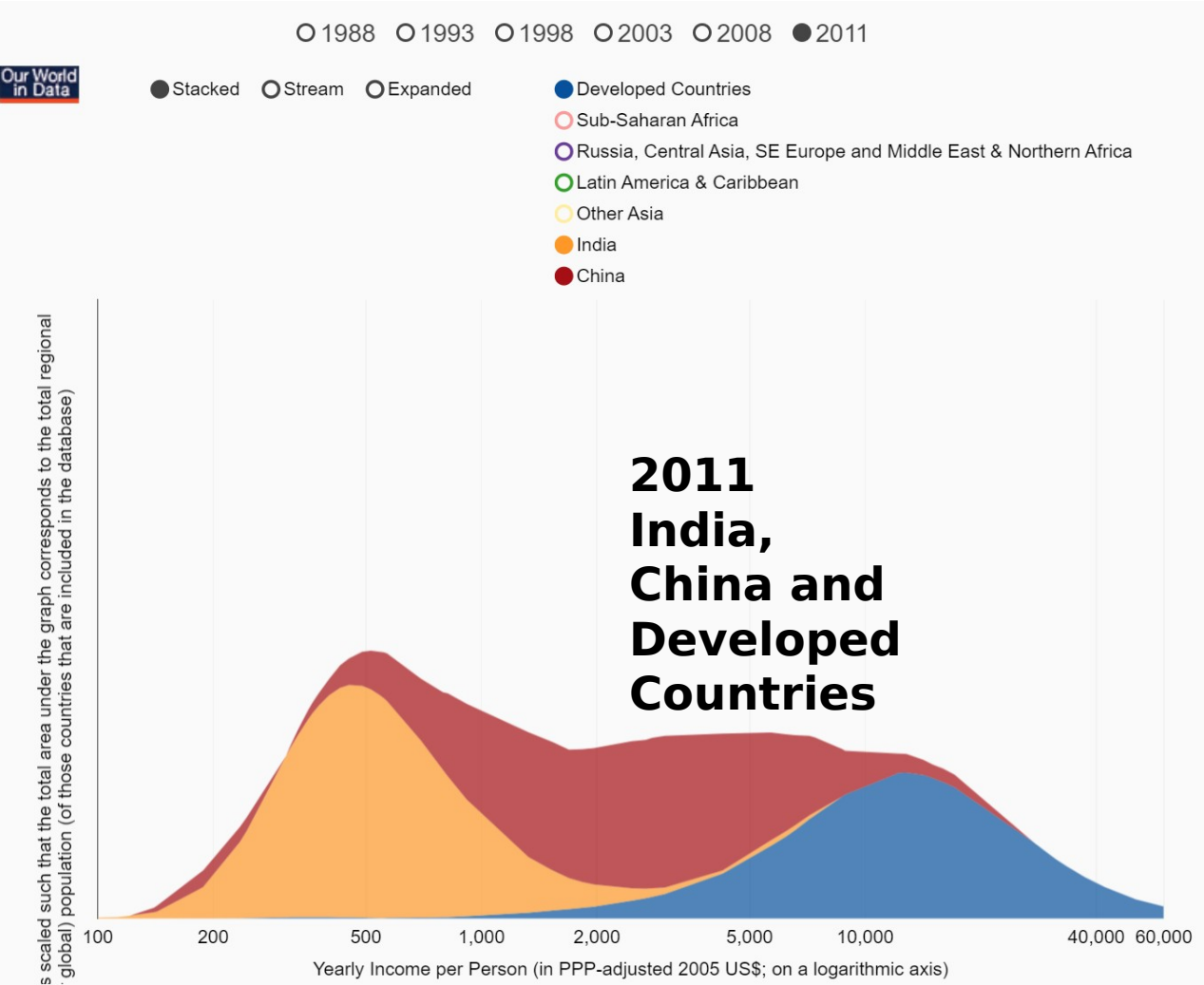
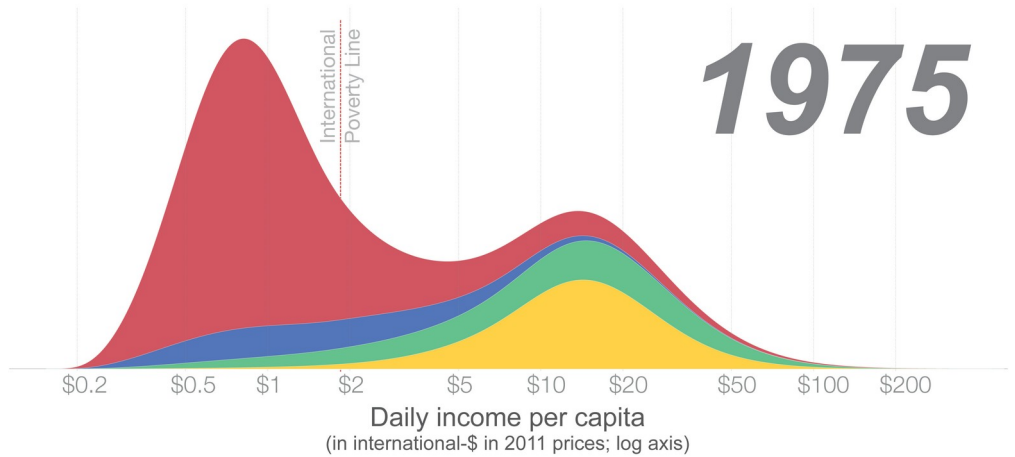
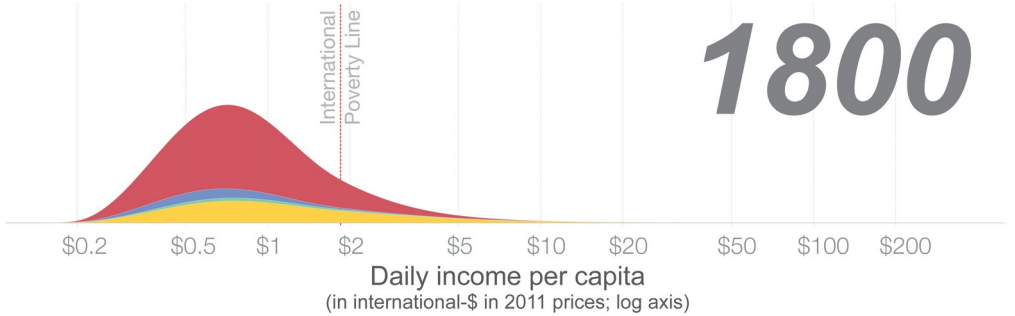
MEASURES OF DISPERSION

Global income distribution in 1800, 1975, and 2015



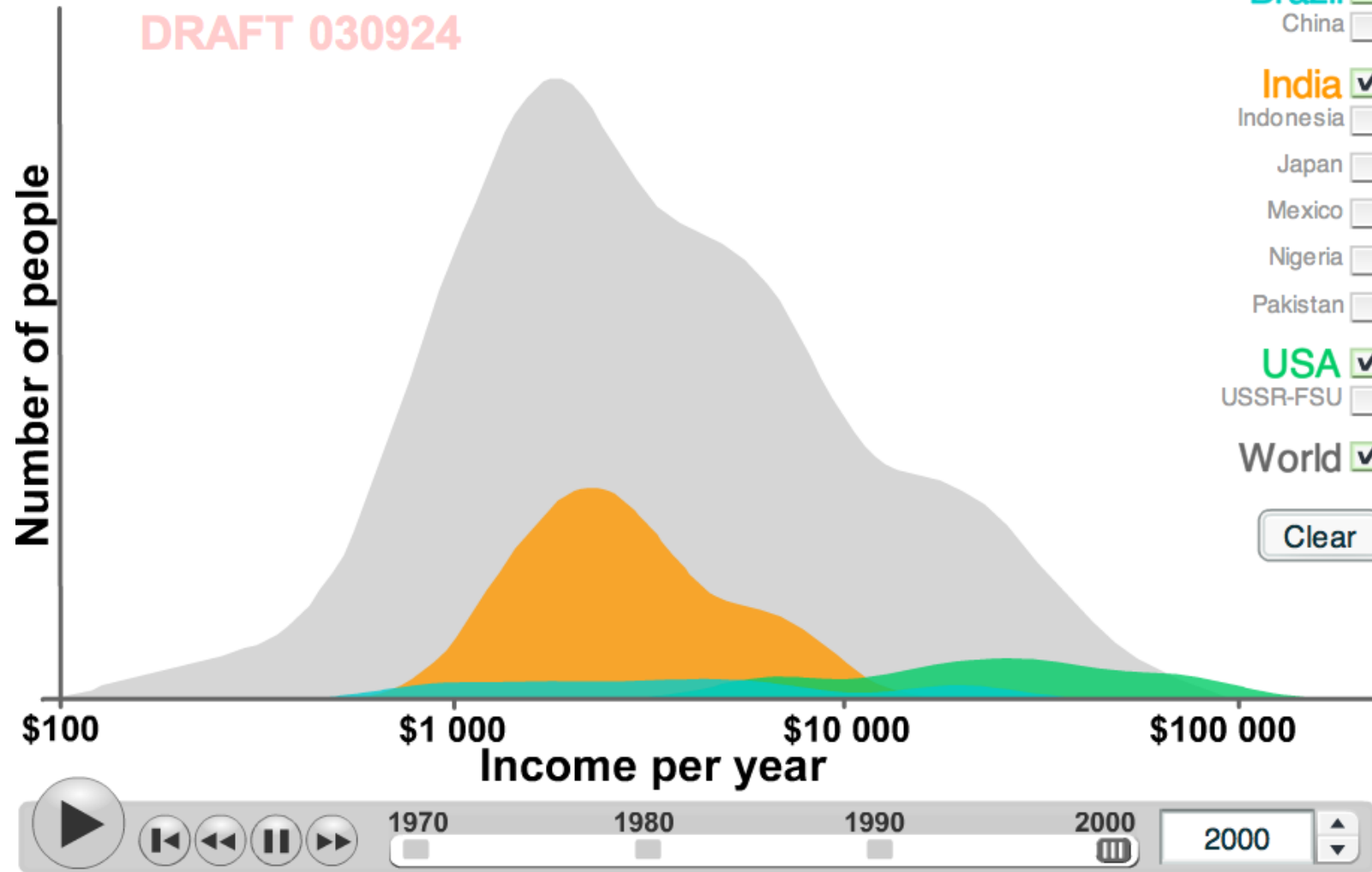
Income is measured by adjusting for price changes over time (inflation) and for price differences between countries (purchasing power parity (PPP) adjustment). These estimates are based on reconstructed National Accounts and within-country inequality measures. Non-market income (e.g. through home production such as subsistence farming) is taken into account. The *International Poverty Line* is set by the *United Nations* and is the the poverty line that defines extreme poverty.

Europe Asia and Pacific Africa North- and South America



<https://ourworldindata.org/global-economic-inequality>

Income distribution 2000



Is there a gap?

Gapminder- app on Income distribution
<https://www.gapminder.org/tag/income-distribution/>

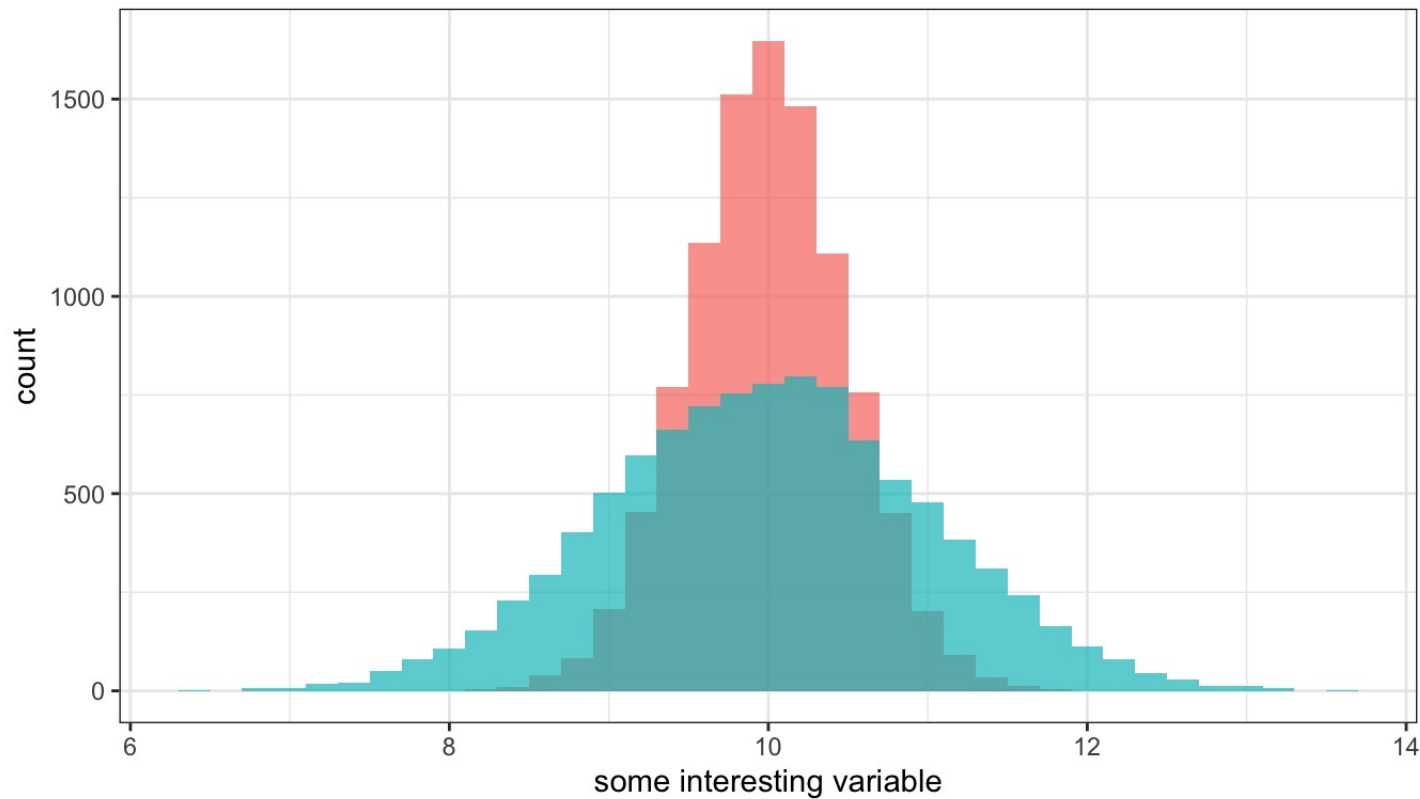


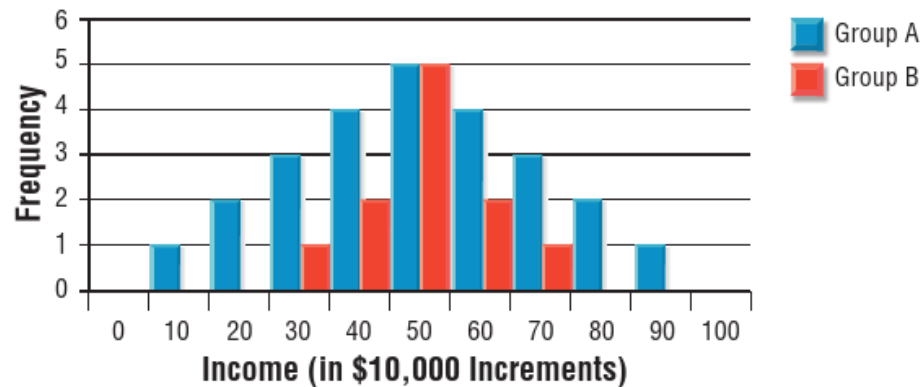
TABLE 1 *Symbols and Formulas*

TERM/SYMBOL	MEANING OF SYMBOL	FORMULA
Measures of dispersion	A group of statistics (range, variance, standard deviation) that is used to describe how spread out cases are across a distribution. They tell us how tightly cases tend to cluster. Used with interval/ratio variables.	
Range	The distance between the case with the highest value and the case with the lowest value	$R = H - L$
Interquartile range	The distance from the third to the first quartile	$Q = Q_3 - Q_1$
Deviation	The distance that a particular case falls from the mean	$D_i = X - \bar{X}$
Mean deviation	The average distance that each case (measured as a positive value) falls from the mean	$D_t = \frac{\sum(X - \bar{X})}{N}$
Sum of squares	The sum of squared deviations from the mean	$ss = \sum(X - \bar{X})^2$
Variance	A type of average. The average of the squared "distances" from the mean.	$s^2 = \frac{\sum(X - \bar{X})^2}{N}$
Standard deviation	A type of average equal to the square root of the variance. It is the square root of the average of the squared "distances" from the mean.	$s = \sqrt{\frac{\sum(X - \bar{X})^2}{N}}$

INCOME DISTRIBUTION – HYPOTHETICAL EX

Now suppose we have two groups of students, Group A and Group B, with six in each group ($N = 6$). **Suppose Group A graduated from one college and Group B graduated from a different college. Five years after**

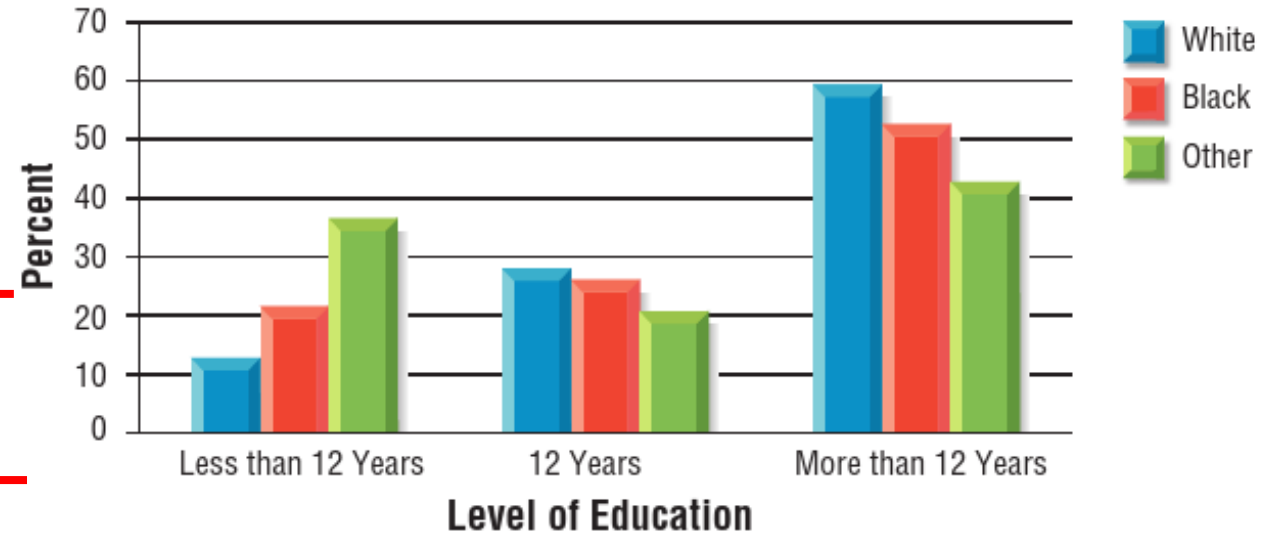
Figure 2 Histograms of Income for Group A and Group B

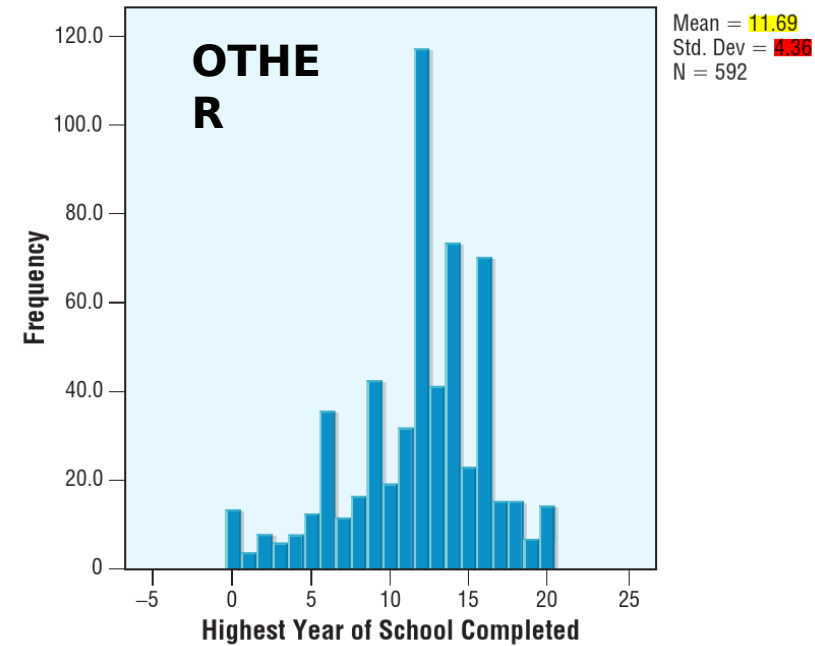
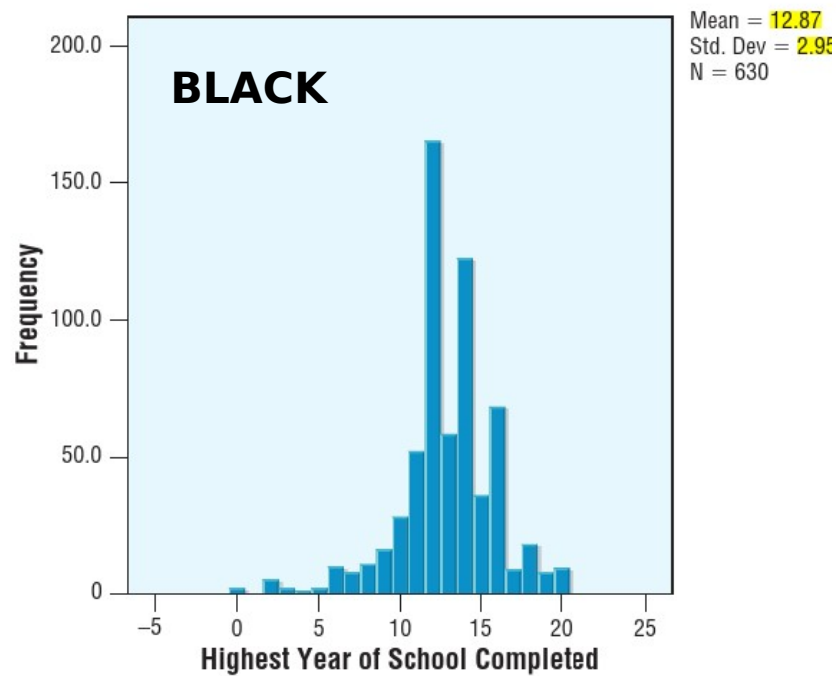
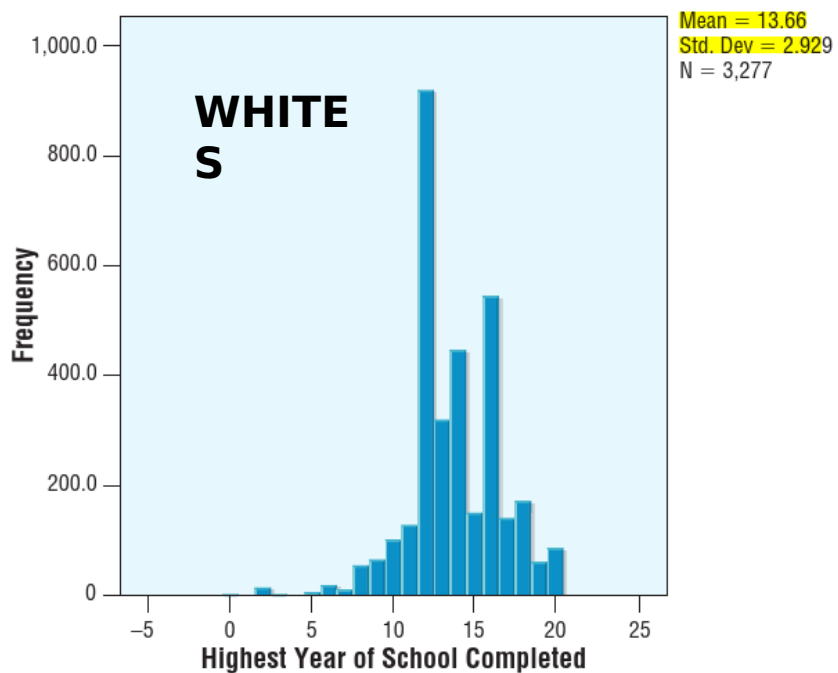


Education Inequality

TABLE 3 *Education by Race*

RACE OF RESPONDENT			FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
WHITE	Valid	Less than 12 Years	428	13.0	13.0	13.0
		12 Years	917	27.9	27.9	41.0
		More than 12 Years	1939	59.0	59.0	100.0
		Total	3284	100.0	100.0	
BLACK	Valid	Less than 12 Years	137	21.6	21.6	21.6
		12 Years	165	26.0	26.0	47.6
		More than 12 Years	332	52.4	52.4	100.0
		Total	634	100.0	100.0	
OTHER	Valid	Less than 12 Years	216	36.5	36.5	36.5
		12 Years	122	20.6	20.6	57.1
		More than 12 Years	254	42.9	42.9	100.0
		Total	592	100.0	100.0	





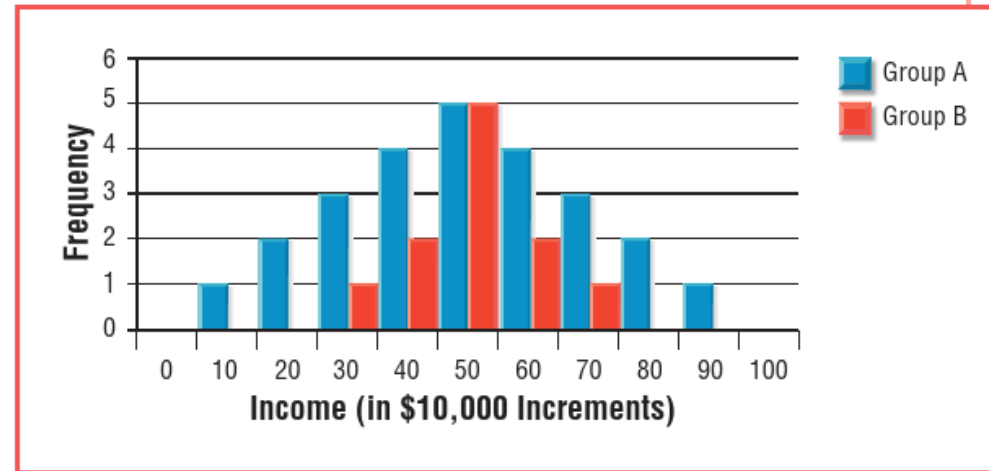
DISPERSION MEASURES

Range The difference between the values of the highest and lowest cases in a distribution

$$R = H - L$$

For group A:

$$R = 90,000 - 10,000 = 80,000$$

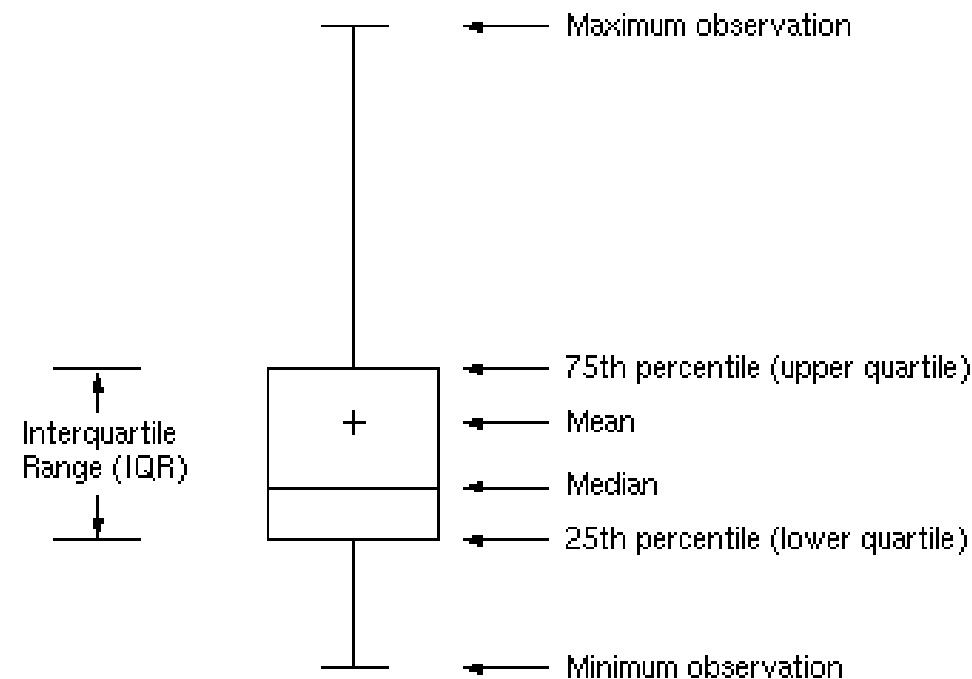
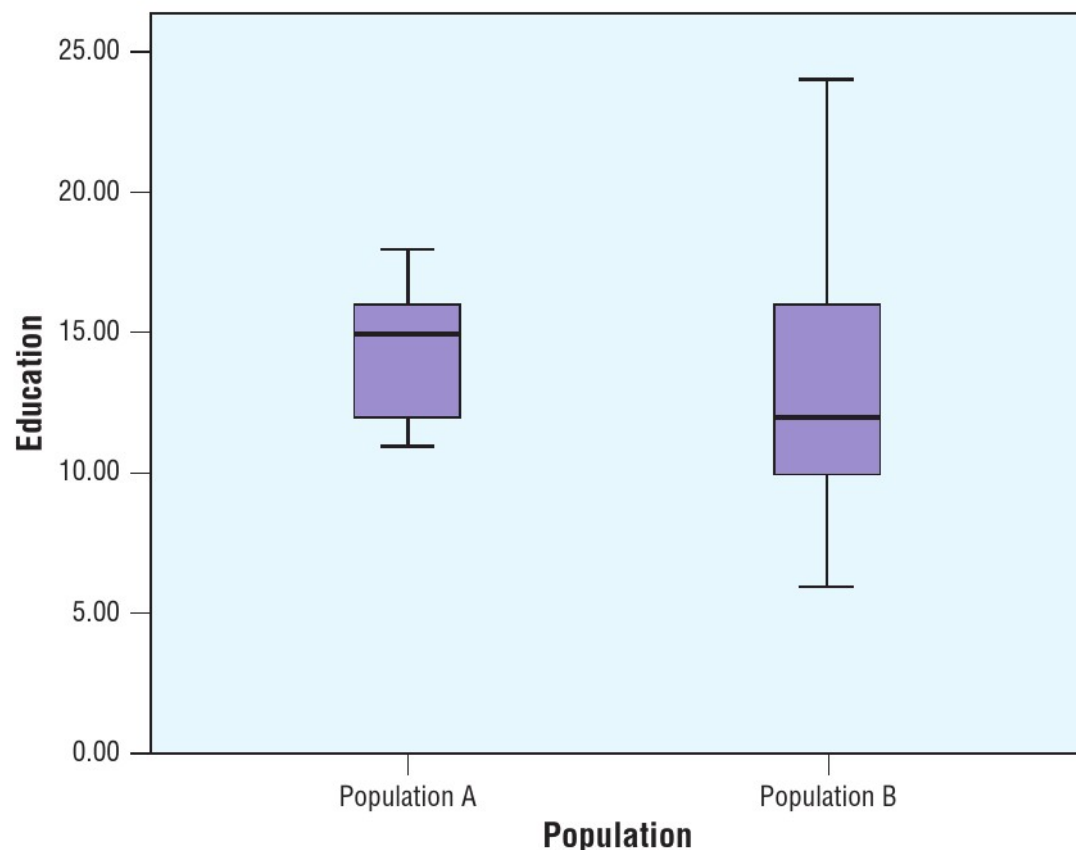


BOX PLOT- QUARTILE RANGE

➤ A given percentile tells you what percent of the distribution is below that number.

➤ The median is the 50th percentile. 50% of the observations are below this value.

➤ Percentiles are sometimes also called **quantiles**



$$IQR = Q3 - Q1$$

IRIS DATASET



Iris Versicolor



Iris Setosa



Iris Virginica



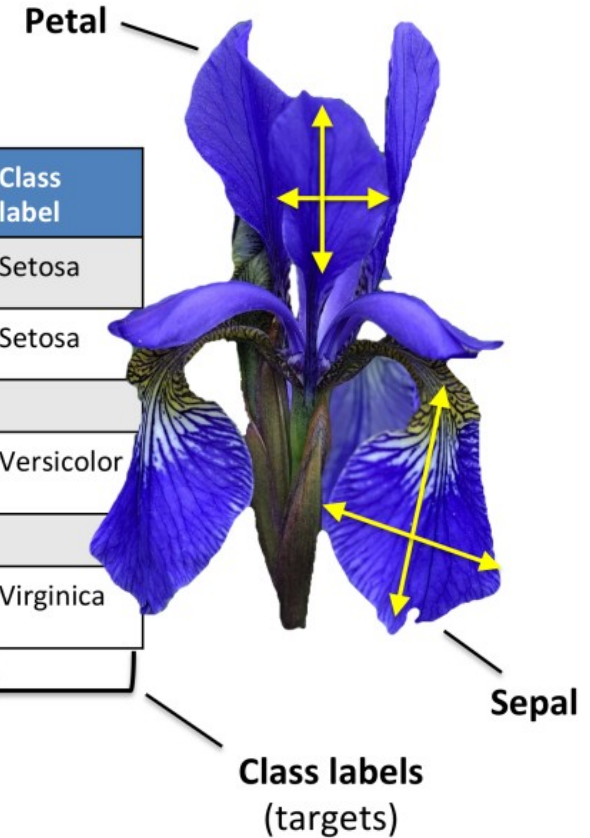
Ronald Fisher (1936)

"The use of multiple measurements in taxonomic problems".

Samples
(instances, observations)

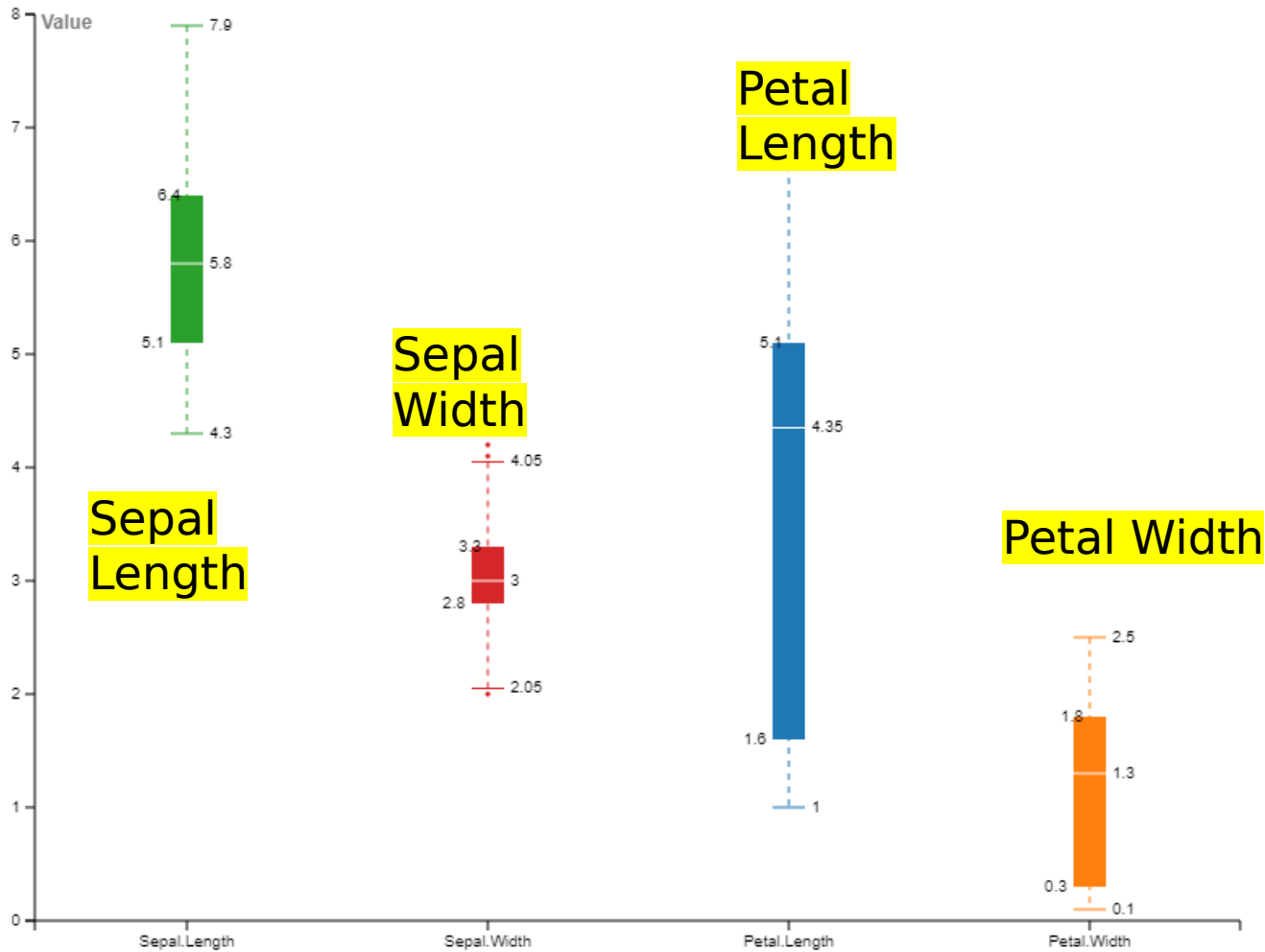
	Sepal length	Sepal width	Petal length	Petal width	Class label
1	5.1	3.5	1.4	0.2	Setosa
2	4.9	3.0	1.4	0.2	Setosa
...					
50	6.4	3.5	4.5	1.2	Versicolor
...					
150	5.9	3.0	5.0	1.8	Virginica

Features
(attributes, measurements, dimensions)



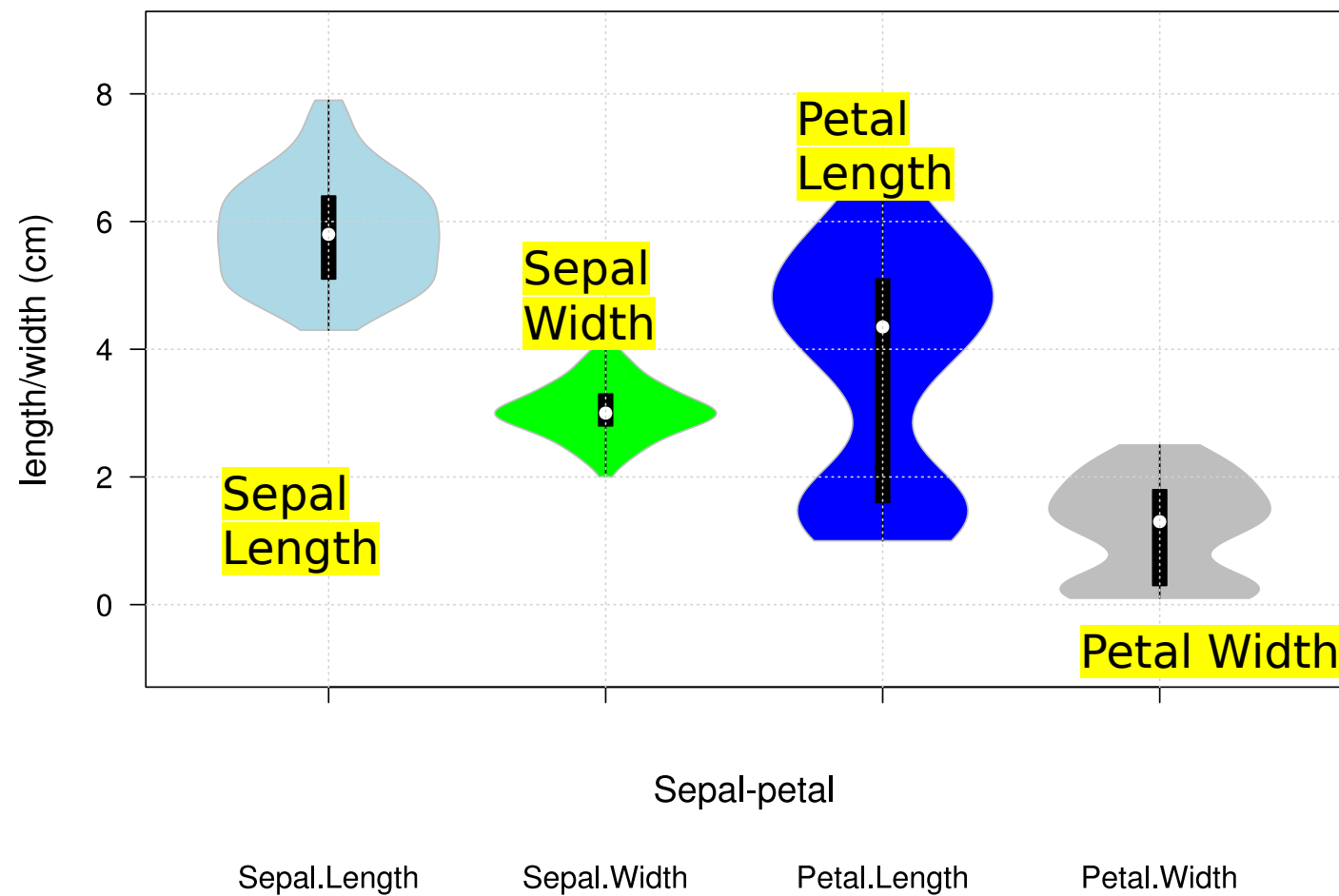
<https://medium.com/analytics-vidhya/exploratory-data-analysis-uni-variate-analysis-of-iris-data-set-690c87a5cd40>

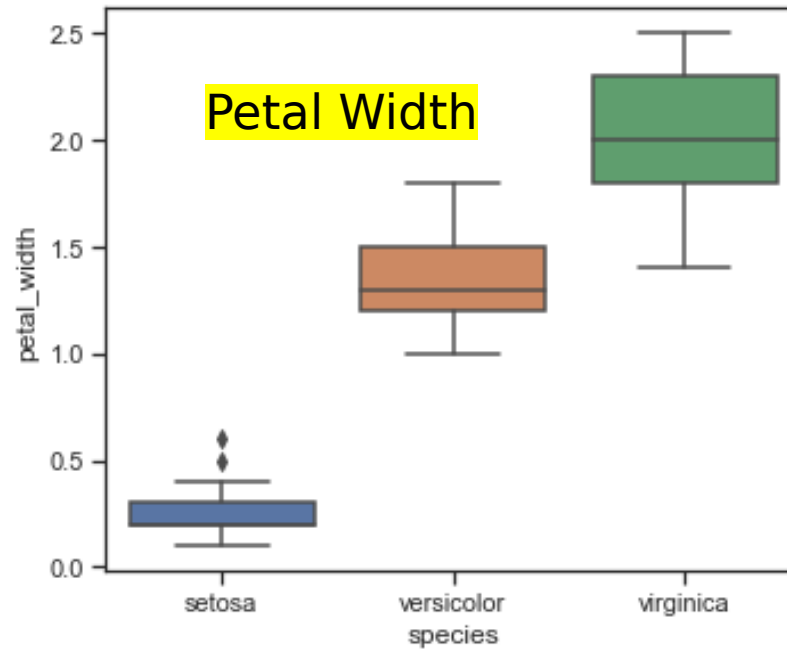
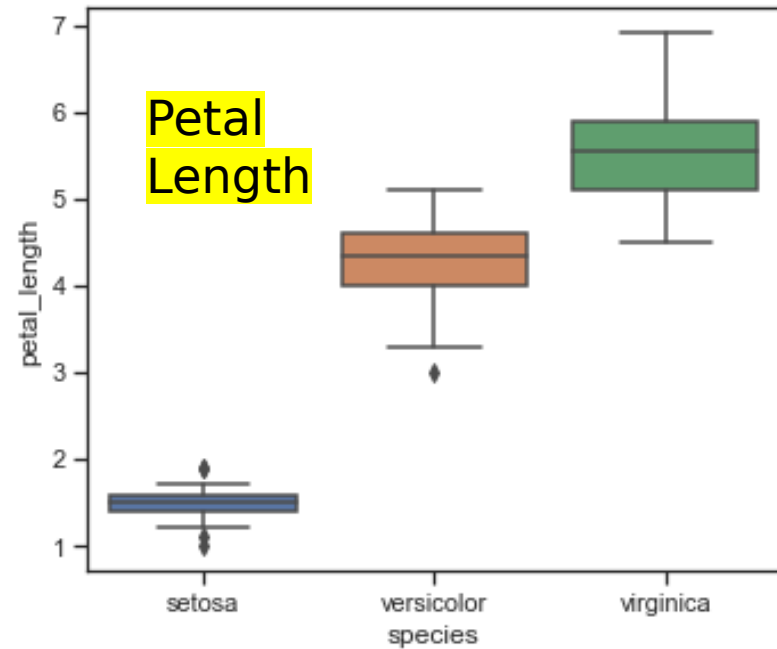
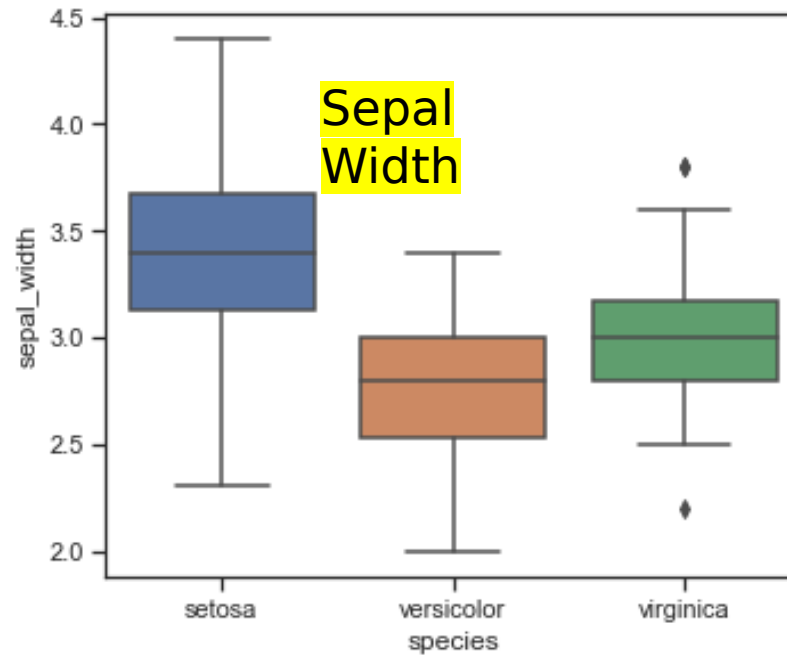
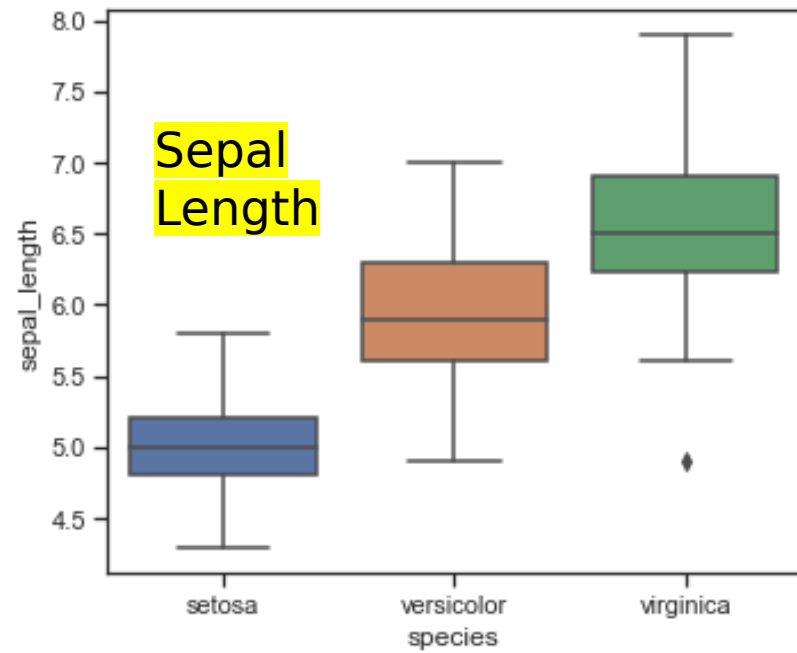
STATISTICS		Sepal.Length	Sepal.Width	Petal.Length	Petal.Width
	Max	7.9	4.4	6.9	2.5
	Min	4.3	2	1	0.1
	RANGE	3.6	2.4	5.9	2.4
	Mean	5.8	3.1	3.8	1.2
	Median	5.8	3.0	4.4	1.3
QUANTILES	0.25	5.1	2.8	1.6	0.3
	0.5	5.8	3	4.35	1.3
	0.75	6.4	3.3	5.1	1.8



VIOLIN PL

IRIS flower data: sepal and petal- length-width distribution



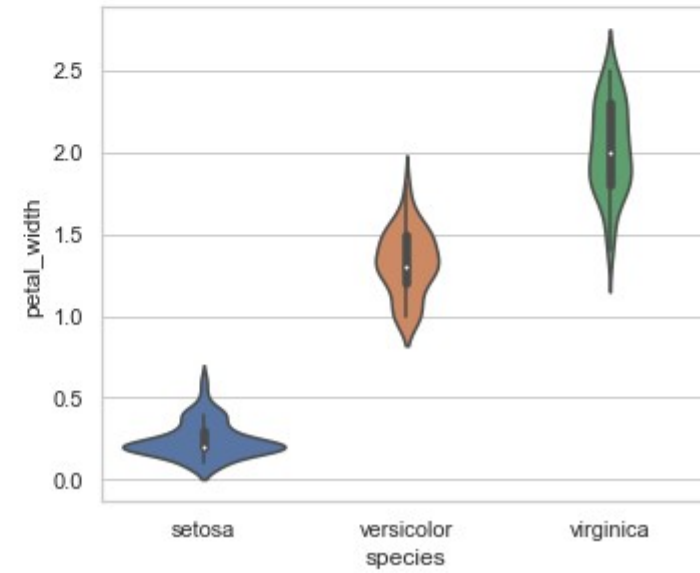
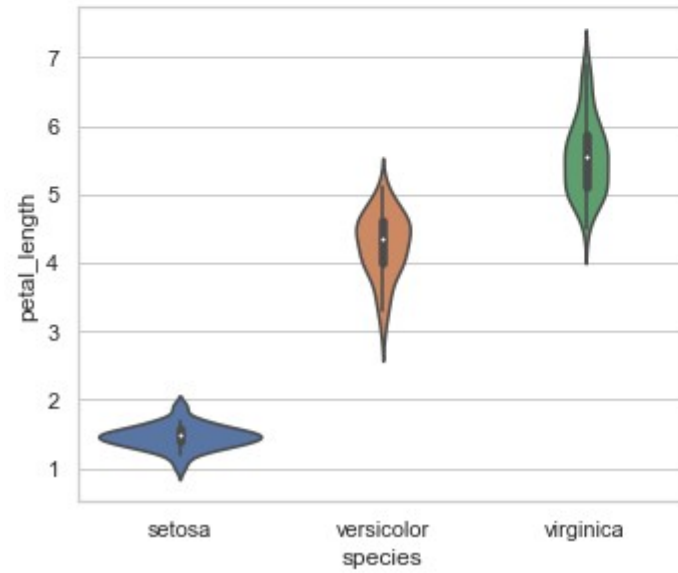
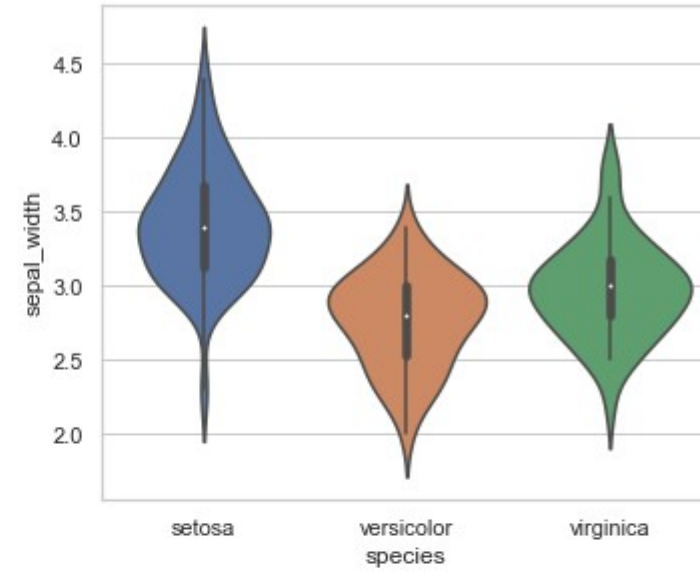
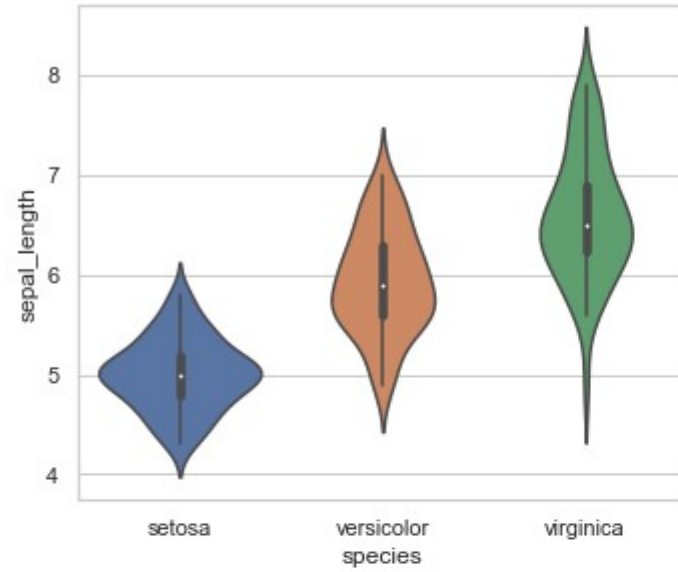


Iris Versicolor

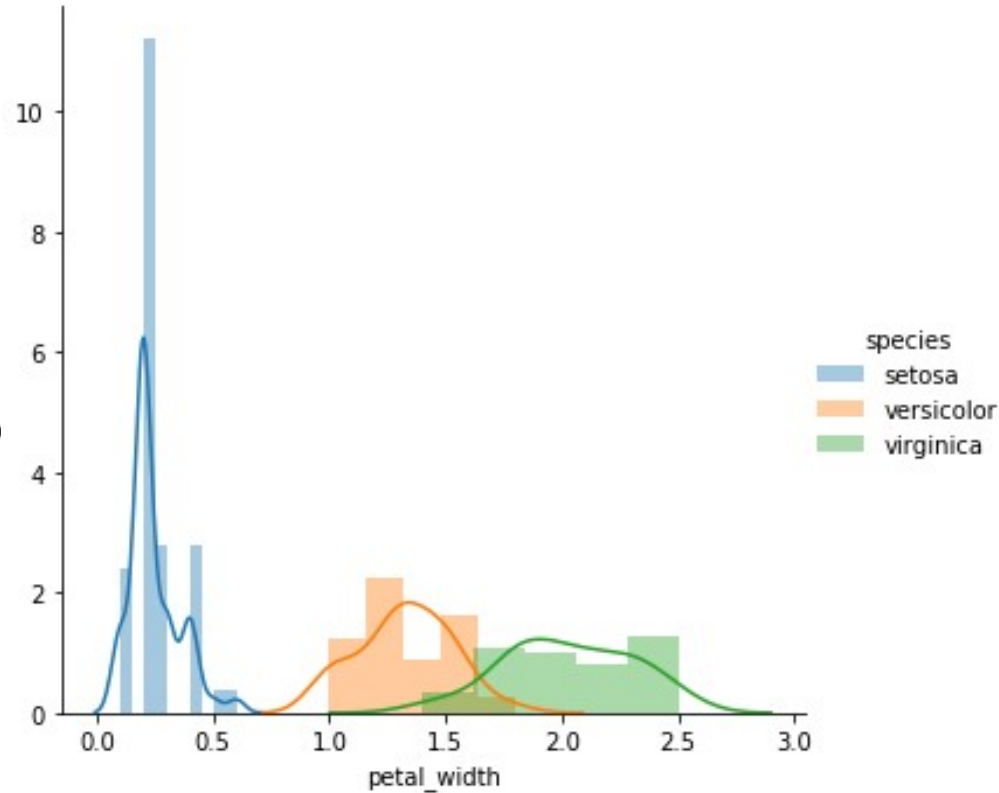
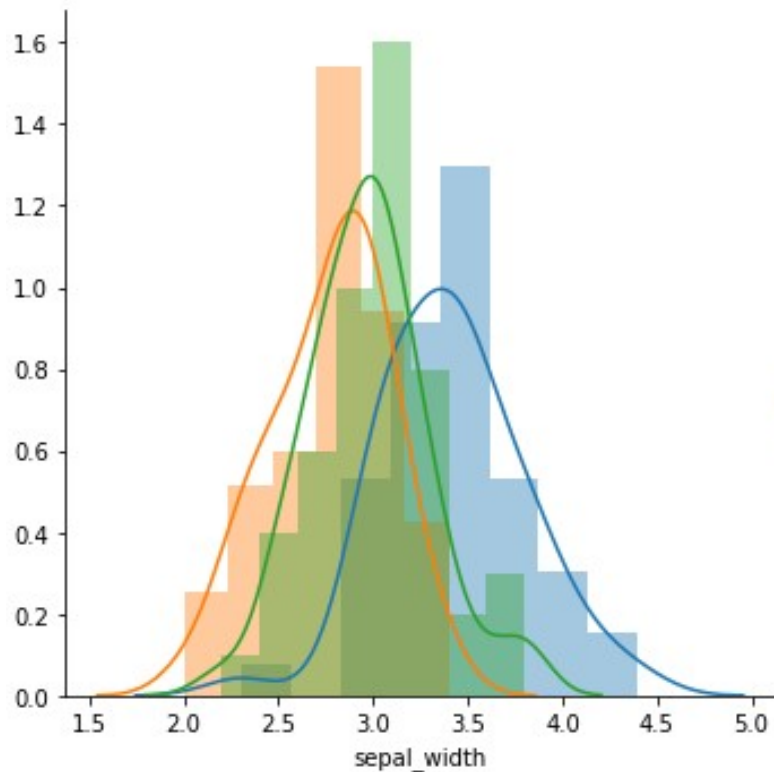
Iris Setosa

Iris Virginica

<https://medium.com/analytics-vidhya/exploratory-data-analysis-uni-variate-analysis-of-iris-data-set-690c87a5cd40>



HISTOGRAM (AND PROBABILITY DISTRIBUTION)



MEAN DEVIATION

$$D_i = |X - \bar{X}|$$

$$D_t = \frac{\Sigma(|X - \bar{X}|)}{N}$$

TABLE 6

RESPONDENT #	RESPONDENT'S EDUCATION	RESPONDENT'S GRANDMOTHER'S EDUCATION	CHANGE IN EDUCATION
1	12	10	2
2	16	12	4
3	12	10	2
4	14	10	4
5	14	12	2
6	12	12	0
7	12	8	4
8	16	12	4
9	16	14	2
10	12	10	2

To begin, we calculate the means as follows:

$$\bar{X} = \frac{\Sigma X}{N} = \frac{2 + 4 + 2 + 4 + 2 + 0 + 4 + 4 + 2 + 2}{10} = \frac{26}{10} = 2.6$$

TABLE 7

CHANGE IN EDUCATION (X)	DEVIATION FROM THE MEAN (X - $\bar{X}$)
2	2 - 2.6 = -.6
4	4 - 2.6 = 1.4
2	2 - 2.6 = -.6
4	4 - 2.6 = 1.4
2	2 - 2.6 = -.6
0	0 - 2.6 = -2.6
4	4 - 2.6 = 1.4
4	4 - 2.6 = 1.4
2	2 - 2.6 = -.6
2	2 - 2.6 = -.6

SQUARED DEVIATION FROM MEAN, AND STANDARD DEVIATION

$$s^2 = \frac{\Sigma(X - \bar{X})^2}{N}$$

$$S^2_{\text{standard}} = \frac{\Sigma(X - \bar{X})^2}{N - 1}$$

TABLE 8

DEVIATION FROM THE MEAN ($X - \bar{X}$)	SQUARED DEVIATIONS FROM THE MEAN
-.6	.36
1.4	1.96
-.6	.36
1.4	1.96
-.6	.36
2.6	6.76
1.4	1.96
1.4	1.96
-.6	.36
-.6	.36
Sum of Squares = 16.4	

$$s^2 = \frac{ss}{N} = \frac{16.4}{10} = 1.6$$

$$s = \sqrt{s^2} = \sqrt{1.6} = 1.26$$

Calculating Variance and Standard Deviation Using Grouped Data

Suppose we sample a group of students living in an off-campus housing facility to better understand the distribution of age. Our data yield the following frequency table in Table 9 and histogram in Figure 8.

TABLE 9 Age of Respondent

		FREQUENCY	PERCENT	VALID PERCENT	CUMULATIVE PERCENT
Valid	18	8	5.1	5.1	5.1
	19	23	14.6	14.6	19.6
	20	27	17.1	17.1	36.7
	21	28	17.7	17.7	54.4
	22	40	25.3	25.3	79.7
	23	32	20.3	20.3	100.0
Total		158	100.0	100.0	

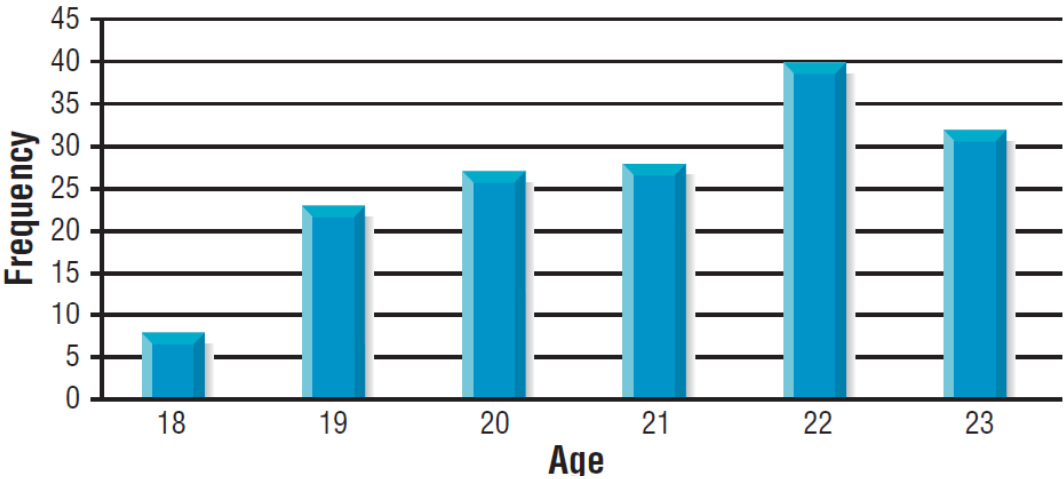


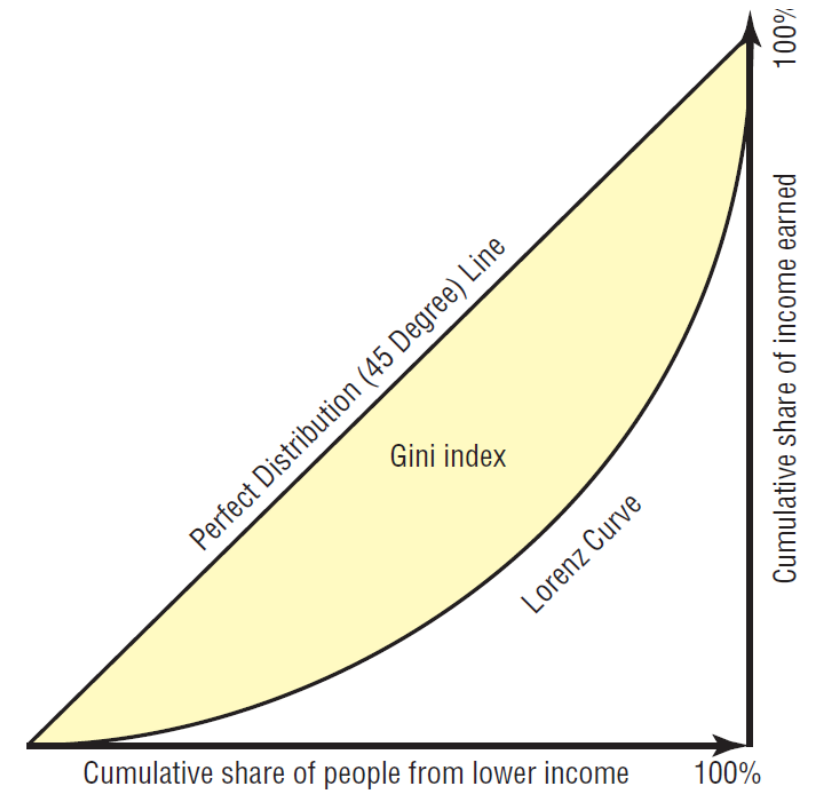
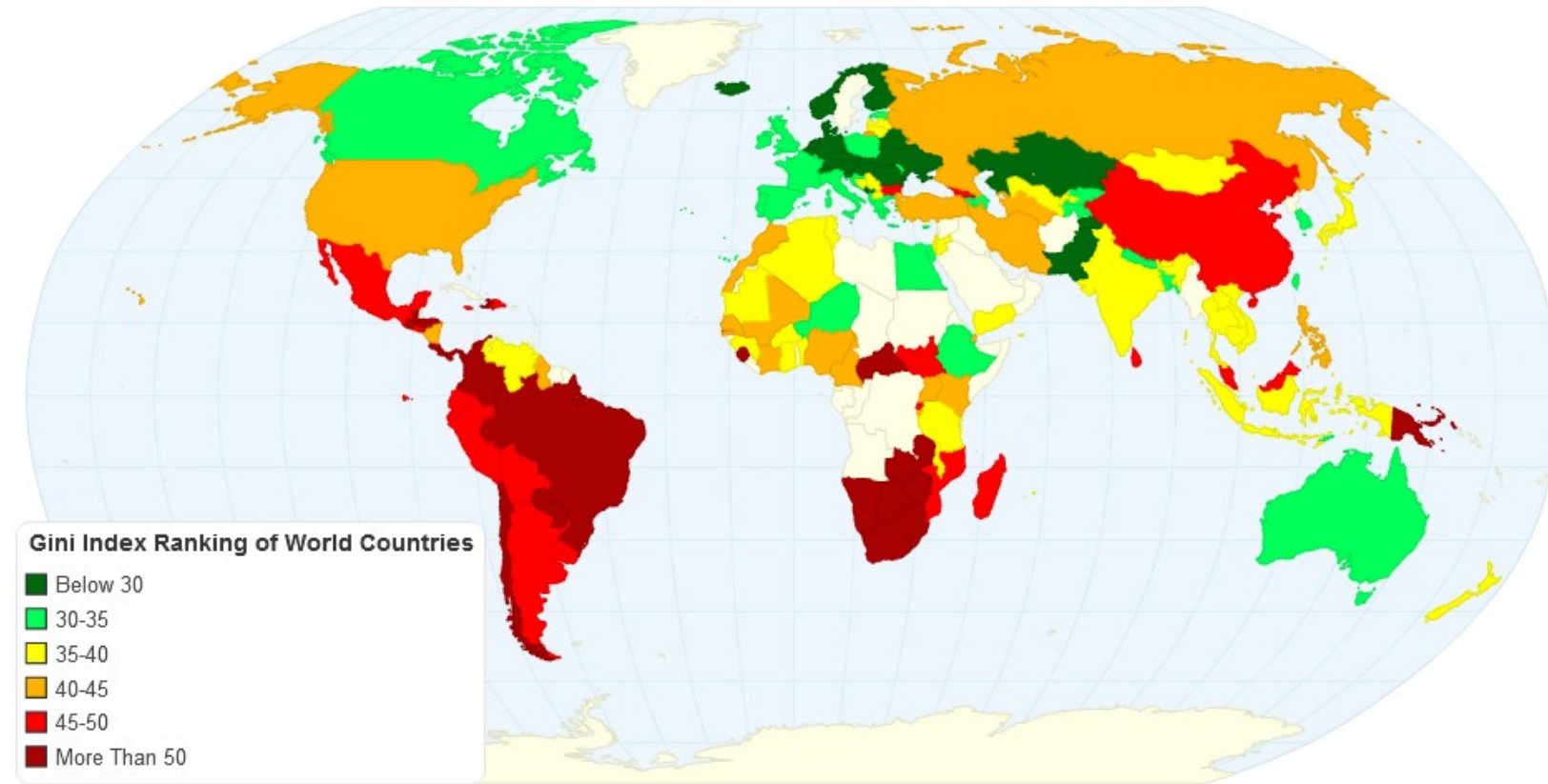
TABLE 10

X	X - $\bar{X}$	(X - $\bar{X}$) ²	(X - $\bar{X}$) ² f
18 (f = 8)	-3	9	9 * 8 = 72
19 (f = 23)	-2	4	4 * 23 = 92
20 (f = 27)	-1	1	1 * 27 = 27
21 (f = 28)	0	0	0 * 28 = 0
22 (f = 40)	1	1	1 * 40 = 40
23 (f = 32)	2	4	4 * 32 = 128
			Sum of Squares = 359

$$s^2 = \frac{\sum(X - \bar{X})^2}{N} = \frac{359}{158} = 2.3$$

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{N}} = \sqrt{\frac{359}{158}} = \sqrt{2.3} = 1.5$$

INCOME INEQUILITY -GINI COEFFICIENT



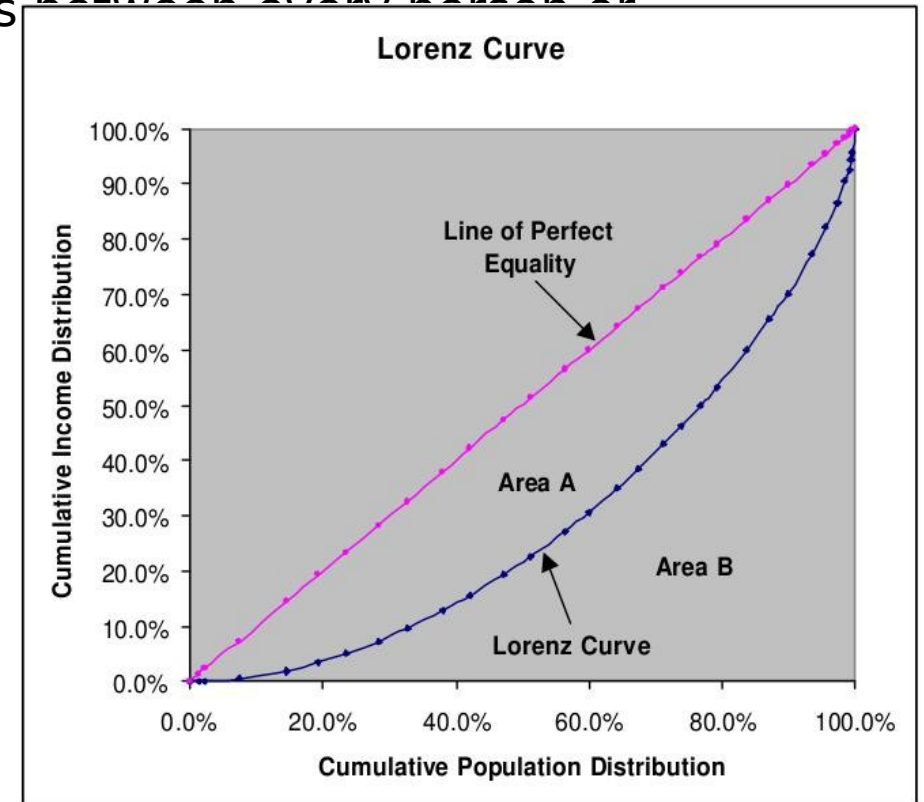
INCOME INEQUALITY

Inequality measures consider the entire distribution (in one or several indicators) and provide a value that summarizes the inequalities between every person or household's achievements of people $i = 1, 2, \dots, n$
Arranged in increasing order $y_1 \leq y_2 \leq y_3, \dots, \leq y_n$

Inequality measure– Gini Coefficient

Lorenz Curve: Plot of Cumulative share of income $L(p)$ versus cumulative proportion of people (p)

$$G = \frac{A}{A + B}$$

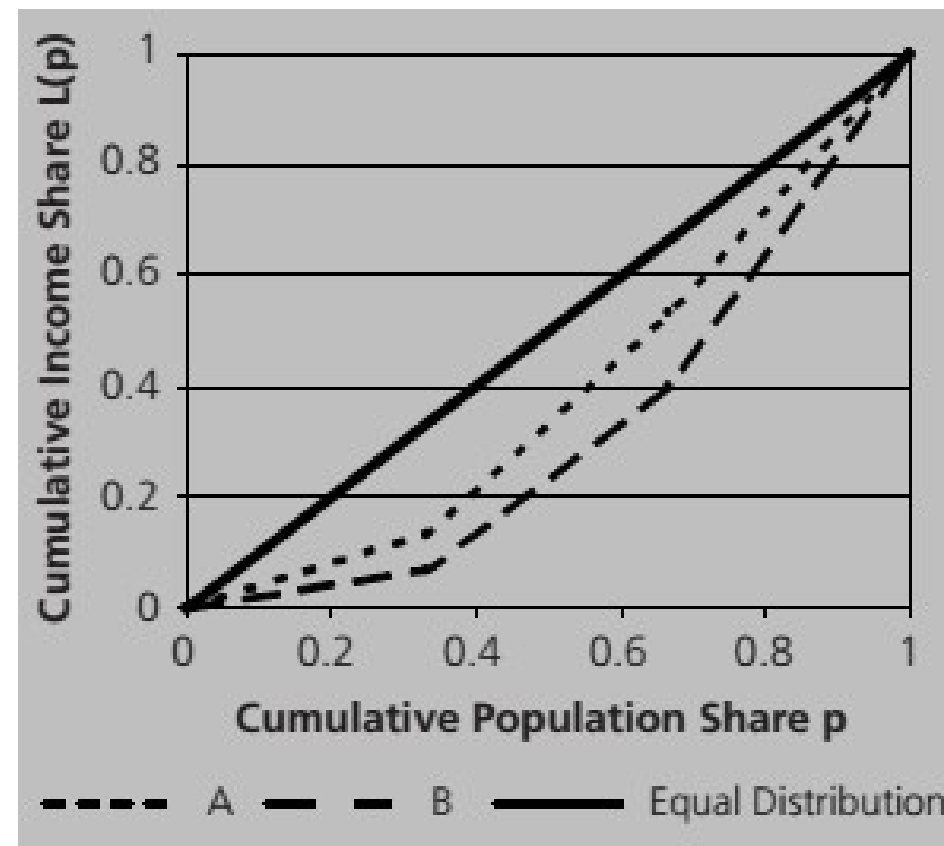


EXAMPLE

Gini Coefficient: The Gini is the ratio of **the area between the diagonal (equal distribution) and the Lorenz curve**, to the entire area in the triangular region below the diagonal.

Persons	State A	State B	State C
1	200	100	500
2	600	500	500
3	700	900	500
Total State	1500	1500	1500

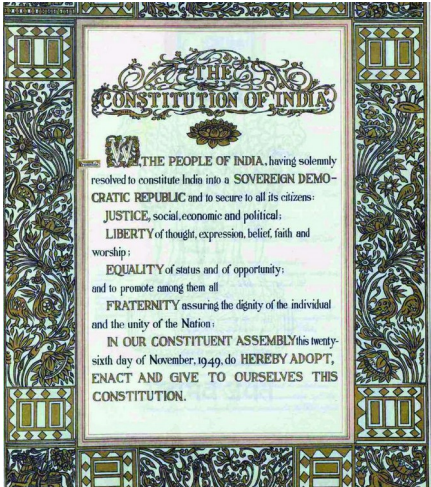
Persons	State A	P	L(p)
1	200	1/3=0.33	200/1500=0.133
2	600	2/3=0.66	800/1500=0.533
3	700	3/3=1	1500/1500=1.0



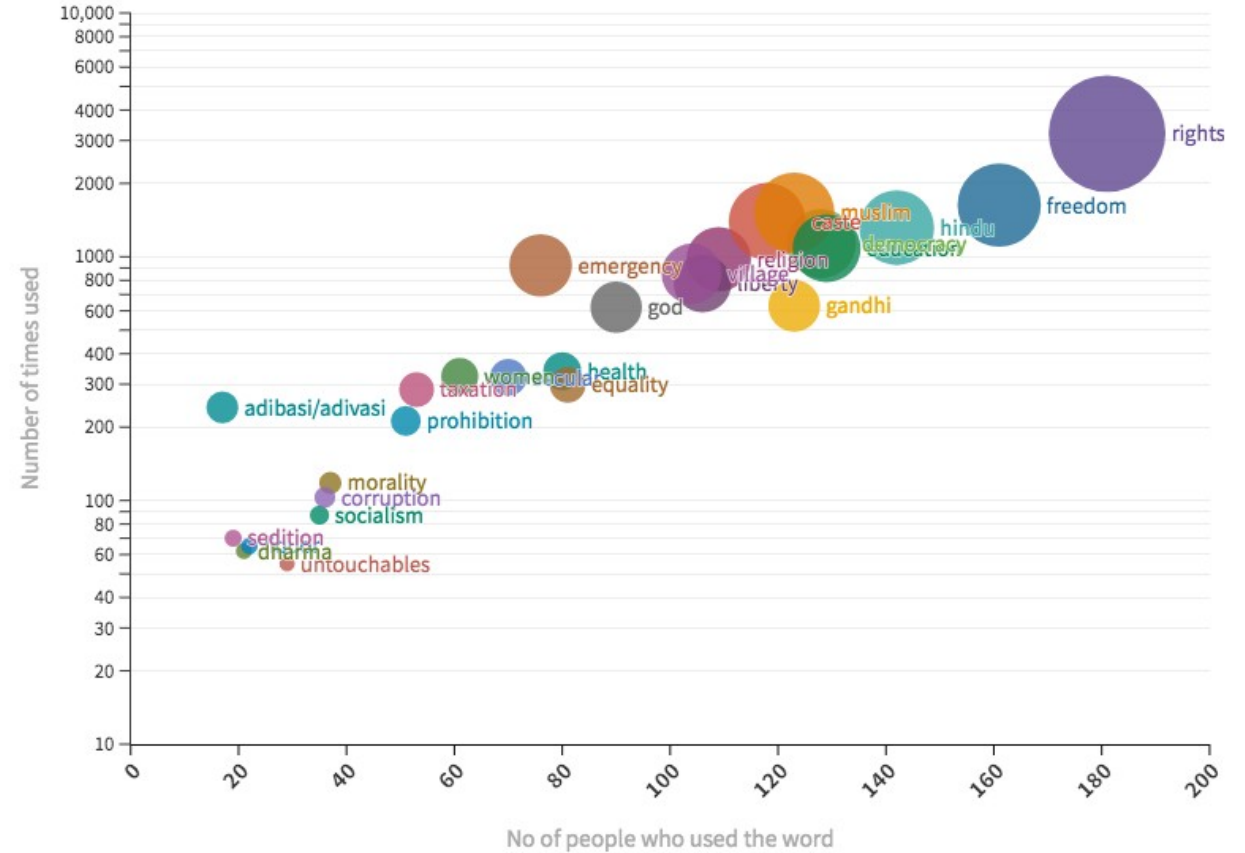
Bottom 33% of the people own only 13% of the total wealth

$$G = \frac{A}{A + B}$$

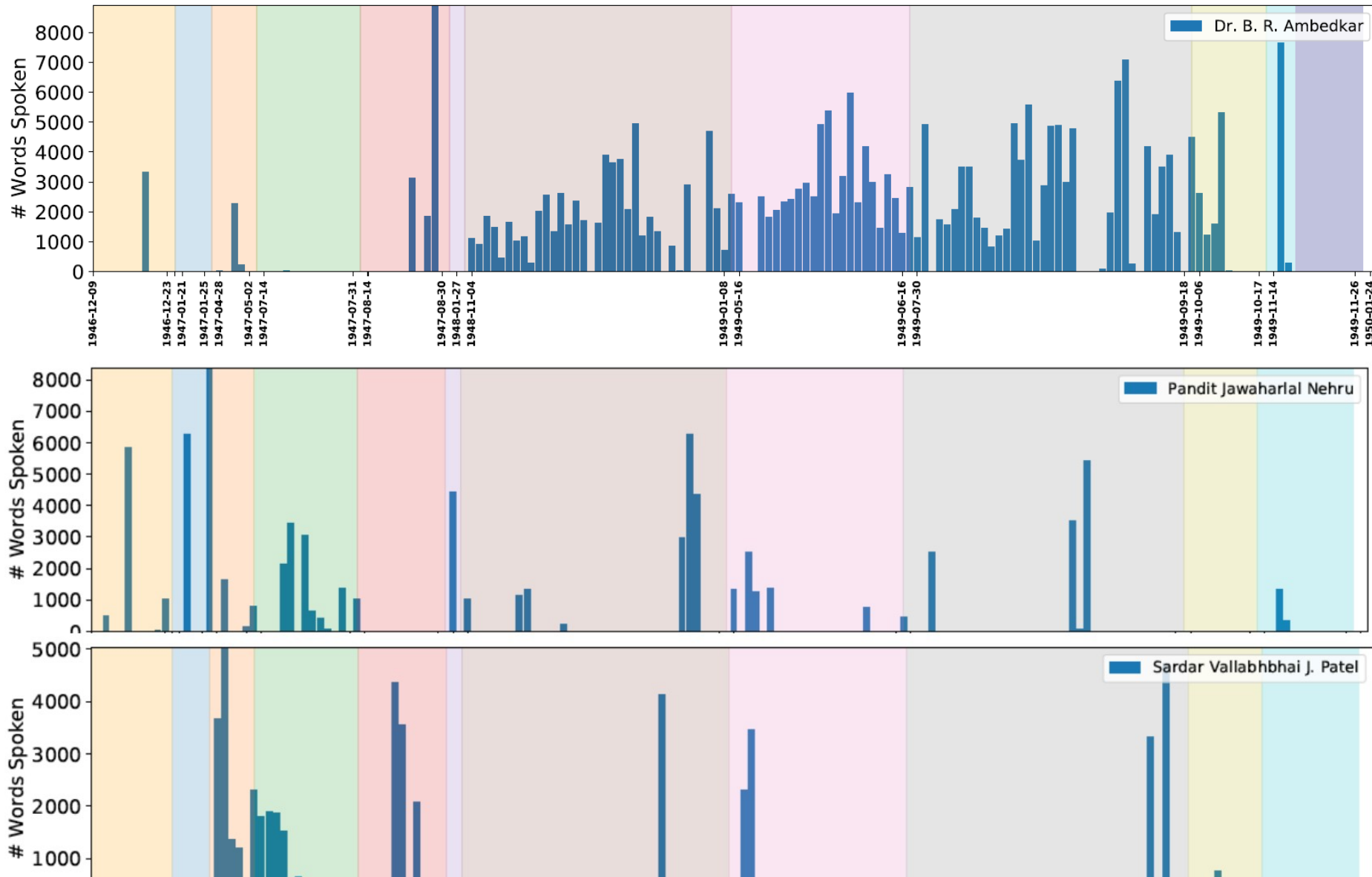
INEQUALITY IN THE CONSTITUENT ASSEMBLY DEBATES



Dec 09 1946- Nov 26, 1949



SPEAKERS AND WORDS



**Shiva Jolad,
Chakresh
Kumar Singh
and Ravi
Vishwakarm**

Constituent Assembly Debates- Words Inequality

Percentile of words versus percentile of people who spoke

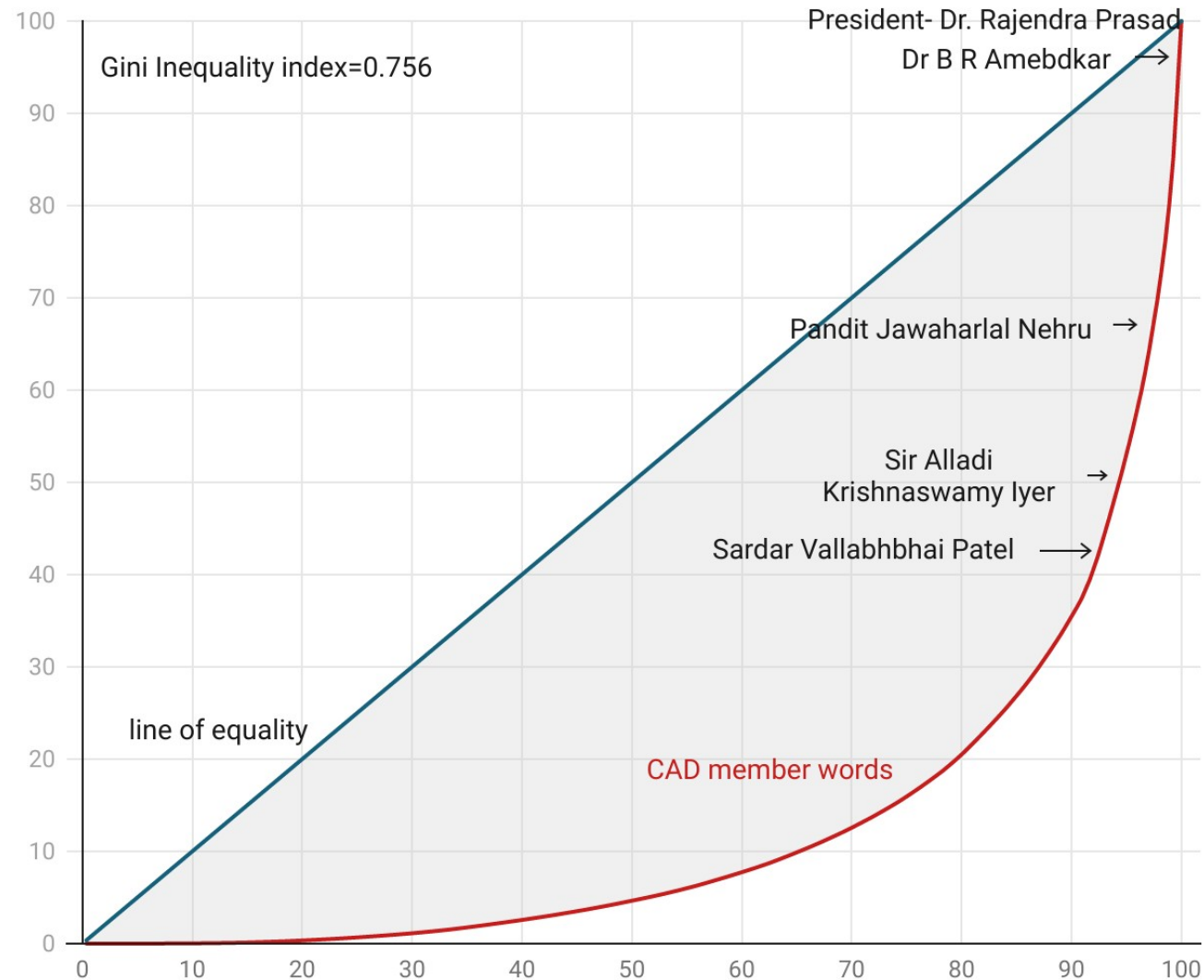


Chart: Shivakumar Jolad • Source: The Constitutional and Civic Citizenship Project- Centre for Law and Policy Research (CLPR) • Created with Datawrapper

What our Constituent Assembly debates reveal of a vital process



Photo: Mint

4 min read . Updated: 25 Nov 2021, 10:35 PM IST

Shivkumar Jolad, Yugank Goyal

FACTFUL VIEW OF WORLD- CLASS QUIZ/VIEW

Factful view of the world- Class questions

Give a honest response - based on your understanding of world. Give the closest option you think (not the most accurate). Your name and email ID is not collected. You will not be graded on this. So need to search for answers on the net or asking your friend.

In all low-income countries across the world today, how many girls finish primary school?

- ☐ 20%
- ☐ 40%
- ☐ 60% or more

In the last 20 years, the proportion of the world population living in extreme poverty has ...?

- ☐ almost doubled
- ☐ remained more or less the same
- ☐ almost halved

<https://forms.gle/rmVVvzEhrVmkRSkK9>